Sample Test 3 Version 2 Answers

April 1, 2025 11:11 AM

Plan: 5, 6, 8, 18

1. Find a polar equation for the curve represented by the Cartesian equation $x^2 + (y-1)^2 = 1$.
(a) $r = \frac{1}{\sin\theta}$ **(b)** $r = 2\sin\theta + 1$ **(c)** $r = 2\sin\theta$ **(d)** $r = 1 - 2\sin\theta$ **(e)** $r = 2\cos\theta$

$x^2+(y-1)^2=1$ $\quad r^2 = 2r\sin(\theta)$

$x^2+y^2-2y+1=1$ $\quad r = 2\sin(\theta)$ (c)

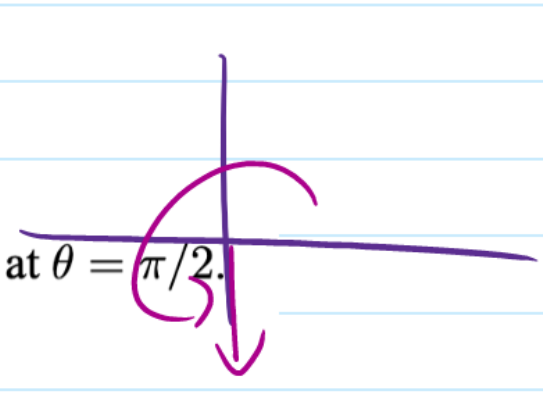
$x^2+y^2 = 2y = r^2$

2. Convert the polar coordinates $(3,5)$ to cartesian coordinates.
(a) $(8, \frac{\pi}{4})$ **(b)** $(4, \frac{\pi}{3})$ **(c)** $(3\cos 5, 3\sin 5)$ **(d)** $(\sqrt{34}, \tan^{-1}\frac{5}{3})$ **(e)** $(3,5)$

$(3,5)$ $\quad r=3$ $\quad \theta=5$

$y = r\sin(\theta) = 3\sin(5)$ $\quad$ (c)

$x = r\cos(\theta) = 3\cos(5)$



3. Find the slope of the tangent line to the curve $r = 3 + \cos 3\theta$ at $\theta = \pi/2$.
(a) -1 **(b)** 0 **(c)** 1 **(d)** -2 **(e)** 2

$$\frac{dy}{dx} = \frac{\frac{dr}{d\theta}\sin\theta + r\cos\theta}{\frac{dr}{d\theta}\cos\theta - r\sin\theta} \qquad \frac{dr}{d\theta} = -\sin(3\theta)(3)$$

$$= \frac{3(1) + \cancel{3(0)}}{\cancel{3(0)} - 3(1)} \qquad = -3\sin\left(\tfrac{3\pi}{2}\right) = 3$$

$= \boxed{-1}$ (a)

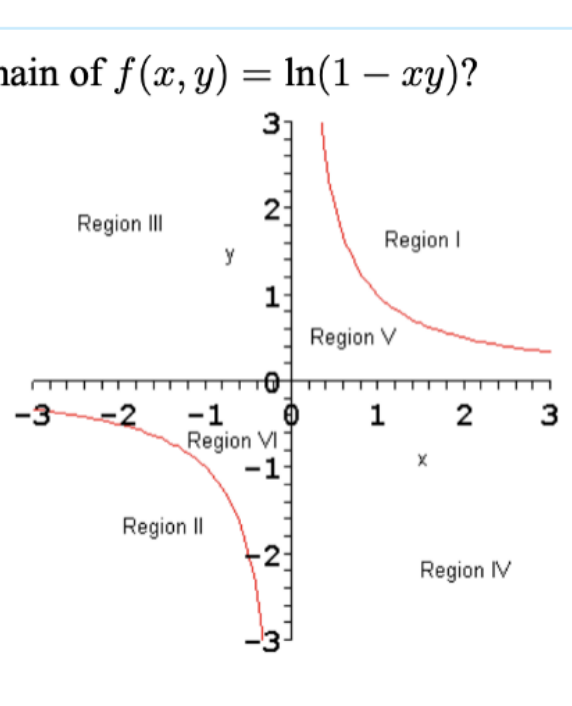
$\sin(\pi/2) = 1$

$\cos(\pi/2) = 0$

$r = 3 + \cos\left(\frac{3\pi}{2}\right) = 3$

4. Which of the regions in the figure below represent the domain of $f(x,y) = \ln(1-xy)$?

(a) Regions I and II not including the curve
(b) Regions II, IV, and V not including the curve
(c) Regions III, IV, V, and VI not including the curve
(d) Regions I, III, and VI not including the curve
(e) Regions II, III, IV, V, and VI not including the curve



$\ln(1-xy)$ to exist: $1-xy>0$ $\quad 1>xy$

Case 1: xy is negative

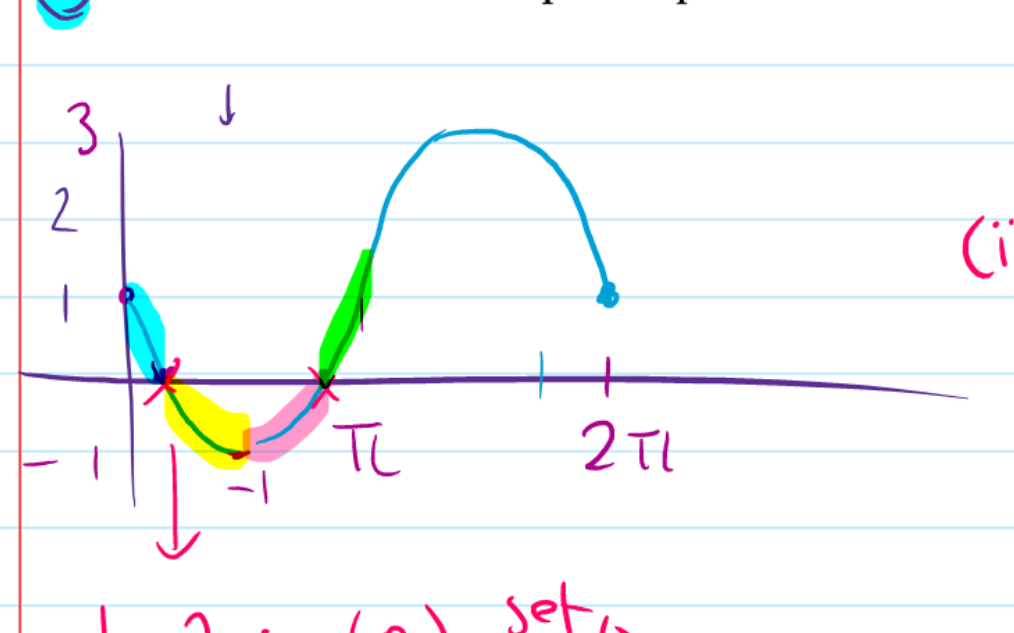
then either $x>0, y<0$ OR $x<0, y>0$

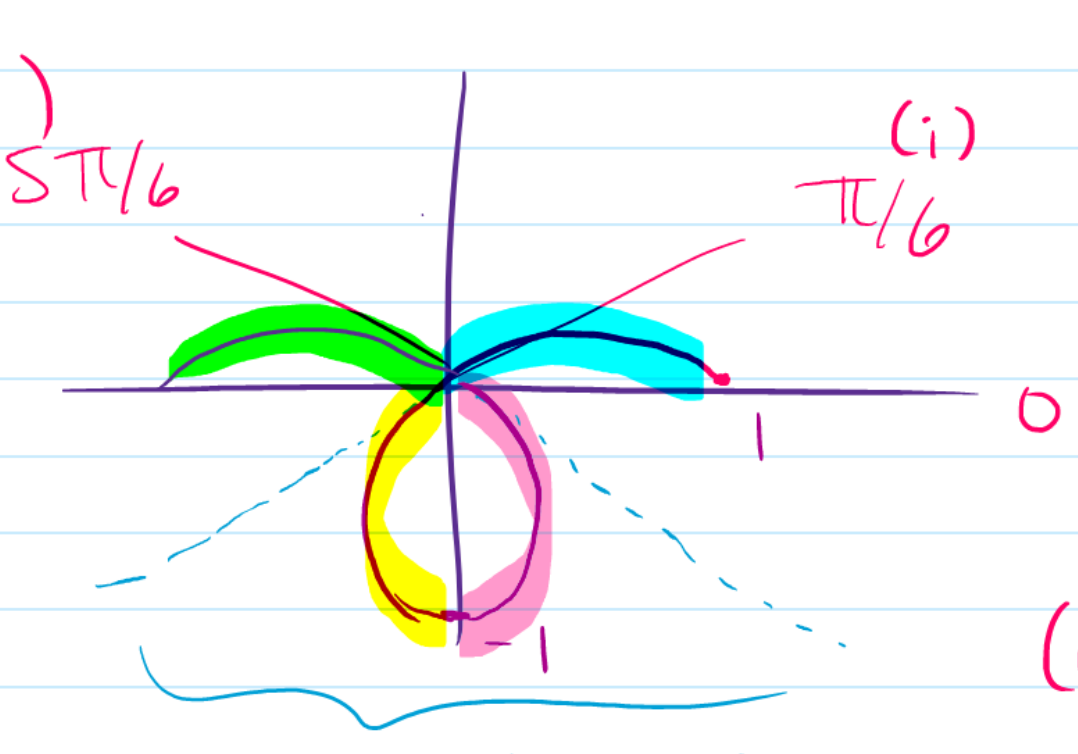
Case 2: xy is positive then

$0<x<1$ and $0<y<1$ OR $0>x>-1$ and $0>y>-1$

By the image: regions III, IV, VI, V (c)

5. Sketch the curve of the polar equation $r = 1 - 2\sin\theta$, $0 \le \theta \le \pi$.





$1 - 2\sin(\theta) \overset{\text{set}}{=} 0$

$\sin\theta = 1/2$

$\theta = \pi/6$ (i)

or $\theta = 5\pi/6$ (ii)

(a)

need to reflect upon this region

6. Let

$$f(x,y) = \begin{cases} \frac{2xy}{x^2+2y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases},$$

and let

$$g(x,y) = \begin{cases} \frac{3xy}{\sqrt{2x^2+y^2}} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}.$$

Which of the above functions are continuous at $(0,0)$?
(a) f only **(b)** g only **(c)** f and g **(d)** neither

first: check f.

$x=0$: $f(0,y) = \frac{0}{0+2y^2} = 0$ $\qquad x=y$: $f(x,x) = \frac{2x^2}{x^2+2x^2} = \frac{2x^2}{3x^2} = \frac{2}{3} \neq 0$

$y=0$: $f(x,0) = \frac{0}{x^2+0} = 0$ $\qquad$ not continuous.

Second: check g.

$x=0$: $g(0,y) = \frac{0}{\sqrt{2x^2}} = 0$ $\qquad x=y$: $g(x,x) = \frac{3x^2}{\sqrt{3x^2}} = \sqrt{3x^2} \xrightarrow{x\to 0} 0$

$y=0$: $g(x,0) = \frac{0}{\sqrt{y^2}} = 0$

$y=mx$: $g(x,mx) = \frac{3x(mx)}{\sqrt{2x^2+(mx)^2}} = \frac{3x^2m}{\sqrt{2x^2+m^2x^2}} = \frac{3x^2m}{\sqrt{x^2(2+m^2)}}$

$= \frac{3|x|^2m}{|x|\sqrt{2+m^2}} = \frac{3|x|m}{\sqrt{2+m^2}} \xrightarrow{x\to 0} 0$

ok: it probably is continuous.

Formal explanation: use squeeze thm.

Note: $x^2 \le 2x^2$ $\qquad$ thus, $\left|\frac{3xy}{\sqrt{2x^2+y^2}}\right| \le 3\left|\frac{\sqrt{2x^2+y^2}\,y}{\sqrt{2x^2+y^2}}\right|$

$\Rightarrow x^2 \le 2x^2+y^2$ $\qquad \le 3|y|$

$\Rightarrow x \le \sqrt{2x^2+y^2}$

thus, $0 \le \left|\frac{3xy}{\sqrt{2x^2+y^2}}\right| \le 3|y|$

limit order thm: $\lim\limits_{(x,y)\to(0,0)} 0 \le \lim\limits_{(x,y)\to(0,0)} \left|\frac{3xy}{\sqrt{2x^2+y^2}}\right| \le \lim\limits_{(x,y)\to(0,0)} 3|y|$

$\Rightarrow 0 \le \lim\limits_{(x,y)\to(0,0)} \left|\frac{3xy}{\sqrt{2x^2+y^2}}\right| \le 0$

$\Rightarrow \lim\limits_{(x,y)\to(0,0)} \left|\frac{3xy}{\sqrt{2x^2+y^2}}\right| = 0$ by squeeze thm.

$\Rightarrow \lim\limits_{(x,y)\to(0,0)} \frac{3xy}{\sqrt{2x^2+y^2}} = 0$ $\qquad$ ans: (b)

7. Let $g(t) = f(t\sin 2t, \cos 2t)$. Suppose that $f_x(0,1) = 2$ and $f_y(0,1) = 3$. Find the value of $g'(0)$.

(a) 0 **(b)** 1 **(c)** 2 **(d)** 3 **(e)** 4

$g(t) = f(t\sin 2t, \cos 2t)$

if $t=0$ then $t\sin 2t = 0$ and $\cos(0) = 1$

$g'(t) = f_x(t\sin 2t, \cos 2t)[\sin 2t + t\cos(2t)(2)]$

$\quad + f_y(t\sin 2t, \cos 2t)[-2\sin(2t)]$

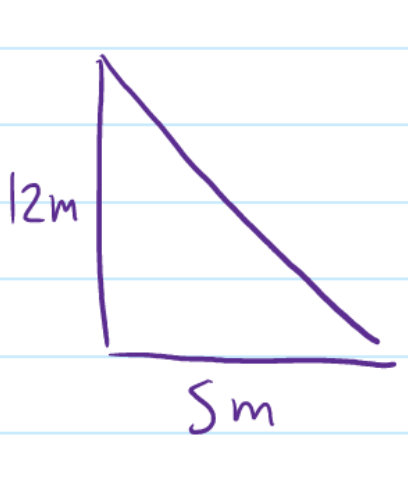
$g'(0) = f_x(0,1)[\sin(0) + 0\cos(0)(2)]$

$\quad + f_y(0,1)[-2\sin(0)]$

$= (2)[0] + (3)[0] = 0$ (a)

8. The two legs of a right triangle are measured as 5 m and 12 m with a possible error in measurement of at most 0.1 cm in each. Use differentials to estimate the maximum error in the calculated value of the length of the hypotenuse.
(a) $\frac{25}{17000}$ **(b)** $\frac{25}{3000}$ **(c)** $\frac{17}{13000}$ **(d)** $\frac{13}{25000}$ **(e)** $\frac{3}{12500}$

note: $0.1\text{cm} \times \frac{1\text{m}}{100\text{cm}} = \frac{0.1}{100} = 0.001$



$h = \sqrt{x^2+y^2}$

$dh = \left(\frac{1}{2}\right)(x^2+y^2)^{-1/2}(2x)\,dx$

$\quad + \left(\frac{1}{2}\right)(x^2+y^2)^{-1/2}(2y)\,dy$

$dh = \frac{x\,dx + y\,dy}{\sqrt{x^2+y^2}} = \frac{5(0.001) + 12(0.001)}{\sqrt{5^2+12^2}}$

$= \frac{17(0.001)}{13} = \frac{1.7}{13(1000)} = \frac{17}{13000}$ (c)

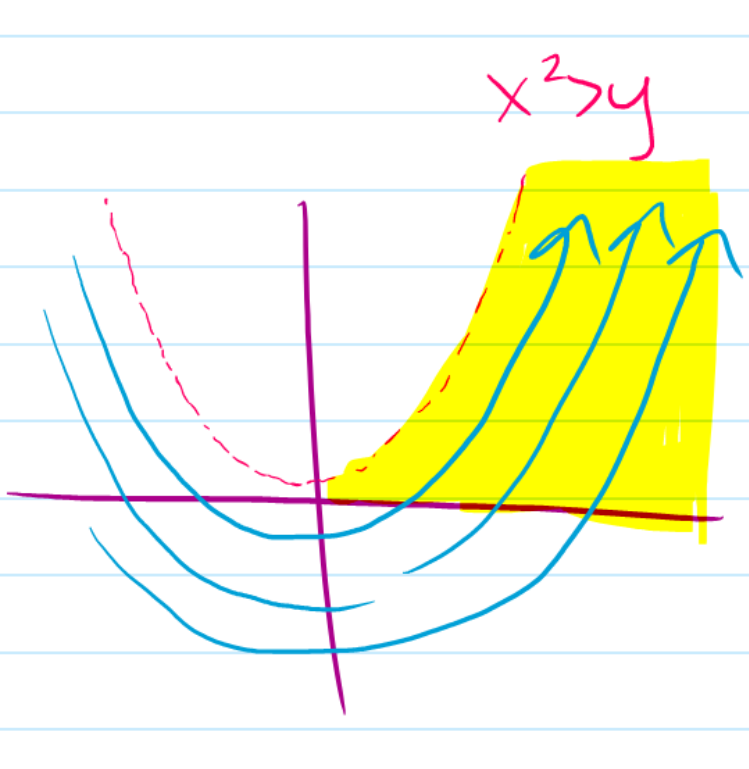
9. Draw a contour plot of $f(x,y) = \ln(x - \sqrt{y})$.

$k = \ln(x-\sqrt{y})$

$e^k = x - \sqrt{y}$

$\sqrt{y} = x - e^k$

$y = (x - \underbrace{e^k}_{>0})^2$



Consider the overlap.

(d)

ALSO, need $x-\sqrt{y}>0$ for $\ln(\cdot)$ to exist

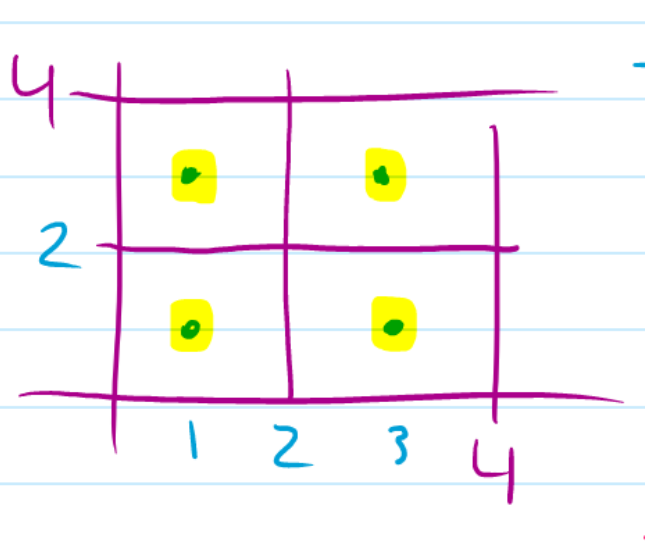
$x>\sqrt{y} \Rightarrow x^2>y$

?!.

10. Matilda wants to use a Riemman sum to approximate the integral

$$\iint_D f(x,y)\,dA$$

where $D = \{(x,y)\,|\,0 \le x \le 4, 0 \le y \le 4\}$. To do so, she divided D into four equal squares and chose a sample point from the middle of each square. The sum of the four pieces was $60+12+12+28 = 112$. This approximation is appropriate if her function $f(x,y)$ was
(a) $4e^{3x^2}y$ **(b)** $6 + \frac{9\sin(\pi x/2)\sin(\pi y/2)}{xy}$ **(c)** $12x^yy^x + 12$
(d) $\ln(x^2+y^2+9) - 4$ **(e)** $6\sqrt{1+x^3}(y^4+3)$



test points: (1,1) (1,3)

(3,1) (3,3)

guess and check...

Test for (a): $4e^{3x^2}y \xrightarrow{(1,1)} 4e^3$ NOT A!

(d): $\ln(x^2+y^2+9) - 4 \xrightarrow{(1,1)} \ln(11) - 4$ NOT D!

(e) $6\sqrt{1+x^3}(y^4+3) \xrightarrow{(1,1)} 6\sqrt{2}(4)$ NOT E! $\qquad \Delta A = (2)(2) = 4$

Try (b): $6 + \frac{9\sin(\frac{\pi x}{2})\sin(\frac{\pi y}{2})}{xy} \xrightarrow{(1,1)} 6 + \frac{9\sin^2(\pi/2)}{(1)} = 6+9 = 15 \xrightarrow{\times 4} 60$

$\xrightarrow{(3,1)} 6 + \frac{9\sin(3\pi/2)\sin(\pi/2)}{3} = 6 + \frac{9(-1)(1)}{3} = 6-3 = 3 \xrightarrow{\times 4} 12$

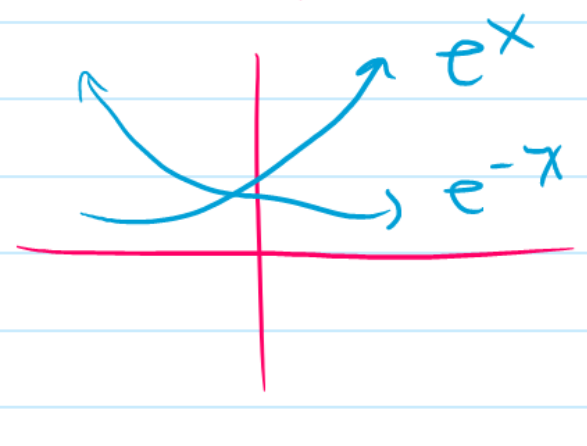
$\xrightarrow{(1,3)}$ similar to (3,1) : $3 \xrightarrow{\times 4} 12$ $\qquad$ HENCE:

$\xrightarrow{(3,3)} 6 + \frac{9\sin^2(3\pi/2)}{9} = 6+1 = 7 \xrightarrow{\times 4} 28$ $\qquad$ matches (b)

11. Sketch the graph of $f(x,y) = \frac{y}{e^x}$. $\quad$ process of elimination!

Note: Keeping x constant, $\frac{y}{e^x}$ is incr. → cannot be (b) or (e)

Keep y constant: e^x is incr $\Rightarrow \frac{1}{e^x} = e^{-x}$ is decr.



ye^{-x} is a decr. function

→ cannot be (c) or (d)

(a) is the graph w/ the above properties

12. Let $z = \frac{x^2}{y}$, $x = rse^t$, $y = re^{st}$. Find $\frac{\partial z}{\partial r}$ when $r=1$, $s=1$, $t=0$.
(a) 0 **(b)** -2 **(c)** -1 **(d)** 2 **(e)** 1

$\frac{\partial z}{\partial r} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial r}$ $\qquad x = rse^t = (1)(1)e^0 = 1$ $\qquad y = re^{st} = (1)e^{(1)(0)} = 1$

$= \frac{2x}{y}se^t - \frac{x^2}{y^2}e^{st}$

$= 2(1)e^0 - \frac{(1)}{(1)}e^0$

$= \boxed{1}$ (e)

13. Find the direction in which $f(x,y,z) = ze^{xy}$ increases most rapidly at the point $(0,1,2)$.
(a) $(1,1,2)$ **(b)** $(2,1,0)$ **(c)** $(1,2,1)$ **(d)** $(1,0,2)$ **(e)** $(2,0,1)$

$\frac{\partial f}{\partial x} = ze^{xy}(y) \xrightarrow{(0,1,2)} (2)e^0(1) = 2$

$\frac{\partial f}{\partial y} = ze^{xy}(x) \xrightarrow{(0,1,2)} 0$

$\frac{\partial f}{\partial z} = e^{xy} \xrightarrow{(0,1,2)} e^0 = 1$

$\nabla f(0,1,2) = 2i + k$

Direction:

$(2,0,1)$ (e)

14. Find the linear approximation of $M(x,y) = (\ln 4x^2) + y^2$ at $(2,4)$, and use it to approximate $M(2.1, 3.8)$.
(a) $3\ln 2 + \frac{27}{2}$ **(b)** $4\ln 2 + \frac{29}{2}$ **(c)** $4\ln 2 + \frac{31}{2}$ **(d)** $3\ln 2 + \frac{29}{2}$ **(e)** $2\ln 4 + \frac{27}{2}$

$L(x,y) \doteq M(a,b) + M_x(a,b)(x-a) + M_y(a,b)(y-b)$

$M(2,4) = \ln(4(2)^2) + 4^2 = \ln(16) + 16$

$M_x = \frac{1}{4x^2}(4(2)(x)) = \frac{2}{x} \xrightarrow{(2,4)} 1$

$M_y = 2y \xrightarrow{(2,4)} 8$

$\doteq \ln(16) + 16 + (1)(2.1-2) + (8)(3.8-4) = 16 + 0.1 + 8(-0.2) + \ln(16)$

$= \ln(4^2) + 14.5 = 2\ln(4) + \frac{29}{2}$ $\qquad 4\ln(2) + \frac{29}{2}$ (b)

Rmrk: $\ln(4^2) = \ln(2^4) = 4\ln(2)$ ← try this version

15. Evaluate

$$\int_0^2\int_0^1 \frac{9x^2y^2}{1+x^3}\,dy\,dx$$

(a) $\ln 3$ **(b)** $\ln 6$ **(c)** $\ln 4$ **(d)** $\ln 9$ **(e)** $\ln 2$

$\int_0^2\int_0^1 \frac{9x^2y^2}{1+x^3}\,dy\,dx = \int_0^2 \frac{9x^2}{1+x^3}\left(\int_0^1 y^2\,dy\right)dx = \int_0^2 \frac{9x^2}{1+x^3}\left(\frac{1}{3}\right)dx$

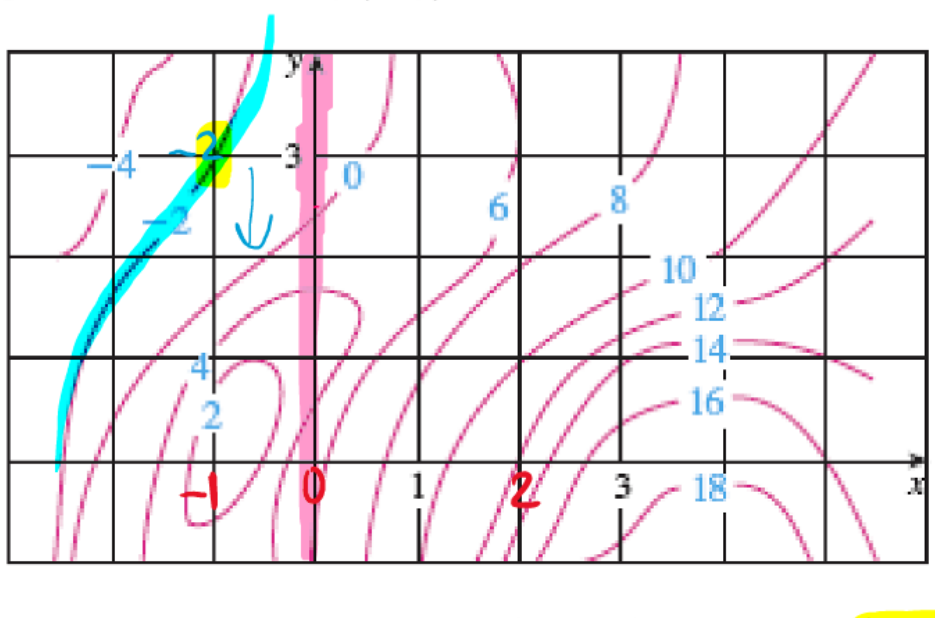
note: $\int_0^1 y^2\,dy = \frac{y^3}{3}\Big|_0^1 = \frac{1}{3}$ $\qquad = 3\int_0^2 \frac{x^2}{1+x^3}\,dx$

$u = 1+x^3$ $\quad du = 3x^2dx \Rightarrow dx = \frac{du}{3x^2}$ $\qquad = \ln|1+x^3|\,\Big|_0^2$

$\cancel{3}\int \frac{\cancel{x^2}}{u}\left(\frac{du}{\cancel{3x^2}}\right) = \int \frac{1}{u}\,du = \ln|u| = \ln|1+x^3|$ $\qquad = \ln(1+8) - \cancel{\ln(1)}^{0}$

$= \ln(9)$ (d)

16. A contour map is given for a function $f(x,y)$.



Based on the contour map, what are the signs of f_x and f_y at the point $(-1,3)$? → kinda have to look at the whole map.

as we go up, the values decr

$\Rightarrow f_y < 0$

as we move more to the right, values incr

$\Rightarrow f_x > 0$

pick (a).

17. Find the equation of the tangent plane to

$$z = \frac{x}{1+xy}$$

at the point $(1, 2, \frac{1}{3})$.

let $f(x,y) = \frac{x}{1+xy}$

$z - z_0 = f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0)$

$f_x = \frac{(1+xy) - x(y)}{(1+xy)^2} = \frac{1}{(1+xy)^2} \xrightarrow{(1,2)} \frac{1}{(1+2)^2} = \frac{1}{9}$

$f_y = \frac{x}{(1+xy)^2}(-1)(x) = \frac{-x^2}{(1+xy)^2} \xrightarrow{(1,2)} \frac{-1}{9}$

thus, $z - \frac{1}{3} = \frac{1}{9}(x-1) + \left(\frac{-1}{9}\right)(y-2)$

$9z - 3 = x - 1 - y + 2$

$9z - 3 = x - y + 1$

$-4 = x - y - 9z$ (e)

18. If

$$e^{xyz} = x + y + z$$

then $\frac{\partial z}{\partial x}$ is equal to
(a) $\frac{-1}{1-e^{-xyz}}$ **(b)** $\frac{yze^{xyz}-1}{1-xye^{xyz}}$ **(c)** $\frac{yze^{xyz}-1-y}{1-xye^{xyz}}$
(d) $yze^{xyz} - 1 - y$ **(e)** $\frac{-xye^{xyz}/z^2 - 1 - y}{1-e^{-xyz}}$

$e^{xyz} = x+y+z \Rightarrow 0 = x+y+z-e^{xyz}$

let $F(x,y,z) = x+y+z-e^{xyz}$

$$\frac{\partial z}{\partial x} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial z}} = -\left(\frac{1-e^{xyz}(yz)}{1-e^{xyz}(xy)}\right) = \frac{e^{xyz}(yz)-1}{1-e^{xyz}(xy)}$$ (b)